

Student: _____
Date: _____
Time: _____

Instructor: Alejandro Melo Ponce
Course: ECO108-Summer 2015
Book: Miller: Economics Today 18e

Assignment: Assignment 4

1. The academic calendar for a university is August 15 through May 15. A professor commits to a contract that binds her to a teaching position at this university for this period. Based on this information, the short run for the professor
- ☐ A. will be the calendar year between January 1 and December 31; any time period longer than this will be long run for her.
 - ☐ B. will be the nine month period between August 15 and May 15; any time period longer than this will be long run for her.
 - ☐ C. will be the time period between August 15 and December 31; any time period longer than this will be long run for her.
 - ☐ D. will be the time period between August 15 of the current year and August 14 of the following year; any time period longer than this will be long run for her.

Answer: B

2. From the perspective of the firm, what is the difference between the short run and the long run?
- ☐ A. The short run is a period of less than a year while the long run is a period of one year or more.
 - ☐ B. In the short run, at least one input is fixed, while in the long run all inputs are variable.
 - ☐ C. There is no difference between the short run and the long-run from the perspective of the firm.
 - ☐ D. In the long run, at least one input is fixed, while in the short-run all inputs are variable.

Answer: B

3. In the long run
- ☐ A. all factors of production are variable.
 - ☐ B. economic profits are always positive.
 - ☐ C. some factors are fixed and some are variable.
 - ☐ D. all factors of production are fixed.

Answer: A

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4. In the long run there
- ☐ A. is only one level of capacity at which the firm can operate.
 - ☐ B. are only variable inputs.
 - ☐ C. are only fixed inputs.
 - ☐ D. are both fixed and variable inputs.

Answer: B

5. The short-run production function for a manufacturer of flash memory drives is shown at the right. Based on this information, calculate the average product at each quantity of labor. (*Your answers should be whole numbers.*)

Input of Labor (workers per week)	Total Output of Flash Memory Drives	Average Product
0	0	
1	30	[
2	54	[
3	72	[
4	84	[
5	95	[
6	96	[

Answers 30
27
24
21
19
16

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6. The short-run production function for a manufacturer of flash memory drives is shown at the right. Based on this information, calculate the marginal product at each quantity of labor. (*Your answers should be whole numbers*).

Input of Labor (workers per week)	Total Output of Flash Memory Drives	Marginal Product
0	0	
1	10	[
2	45	[
3	70	[
4	90	[
5	100	[
6	105	[

Answers 10
 35
 25
 20
 10
 5

7. If total product is increasing at a decreasing rate, then marginal product is

- ☐ A. constant.
☐ B. increasing at a decreasing rate.
☐ C. increasing at an increasing rate.
☐ D. decreasing.

Answer: D

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8. If a firm hires an additional worker and discovers that its total output has fallen, then it must be true that

- ☐ A. marginal product is minimized and marginal cost is maximized.
- ☐ B. average total cost is negative.
- ☐ C. marginal cost is negative.
- ☐ D. marginal product is negative.

Answer: D

9. The law of diminishing marginal returns shows the relationship between

- ☐ A. inputs and outputs for a firm in the short run.
- ☐ B. short run inputs and long run outputs for a firm.
- ☐ C. inputs and outputs for a firm in the long run.
- ☐ D. accounting and economic profits.

Answer: A

10. In the short run, if a firm continues to add workers, marginal product must begin to diminish because

- ☐ A. average product must be declining.
- ☐ B. new workers are less qualified than previous workers.
- ☐ C. each worker has less capital to work with.
- ☐ D. the saturation point has been achieved.

Answer: C

11. During the previous month, a firm produced 400 tablet devices and its total costs were \$21,200. Just before the firm produced its last tablet device in the previous month, its total costs were \$21,120.

The marginal cost incurred by the firm in producing the final tablet device that month was \$. (Enter your response as a whole number.)

Answer: 80

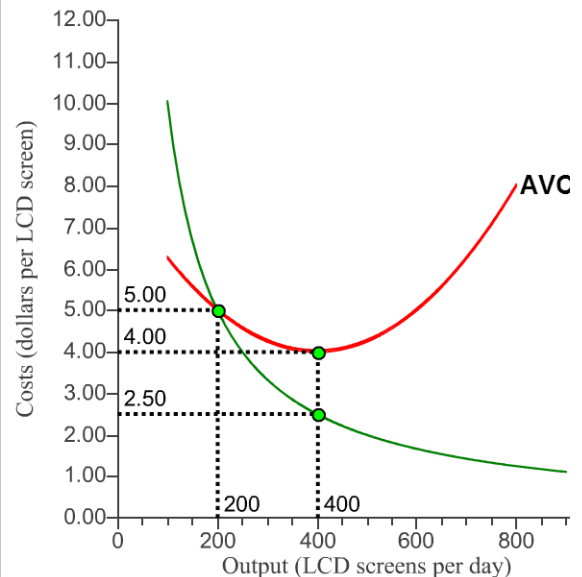
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12. The diagram to the right displays short-run cost curves for a facility that produces liquid crystal display (LCD) screens for cell phones.

- a. What are the daily total fixed costs of producing 400 LCD screens? \$ (Round your answer to the nearest penny.)
- b. What are the total variable costs of producing 400 LCD screens per day? \$ (Round your answer to the nearest penny.)
- c. What are the total costs of producing 400 LCD screens per day? \$ (Round your answer to the nearest penny.)
- d. What is the average total cost of producing 400 LCD screens? \$ (Round your answer to the nearest penny.)



Answers 1,000
 1,600.00
 2,600.00
 6.50

13. At its current short-run level of production, a firm's average variable costs equal \$30 per unit, and its average fixed costs equal \$30 per unit. Its total costs at this production level equal \$1,200.

What is the firm's current output level? units.

What are its total variable costs at this output level? \$.

What are its total fixed costs? \$.

Answers 20
 600
 600

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14. In the short run, a firm's total costs of producing the hundredth unit of output equal \$10,000. If it produces one more unit, its total costs will increase to \$10,200.

What is the marginal cost of the 101st unit of output? \$ (Round your answer to the nearest cent.)

What is the firm's average total cost of producing 100 units? \$ (Round your answer to the nearest cent.)

What is the firm's average total cost of producing 101 units? \$ (Round your answer to the nearest cent.)

Answers 200.00

100.00

100.99

-
15. Suppose that a company currently employs 2,000 workers and produces 2 million units of output per month. Labor is its only variable input, and the company pays each worker the same monthly wage. The company's current total variable costs equal \$4 million.

What are the average variable costs at this firm's current output level? \$ (Enter your response as a whole number).

What is the average product of labor? (Enter your response as a whole number).

What monthly wage does the firm pay each worker? \$ (Enter your response as a whole number).

Answers 2.00

1,000

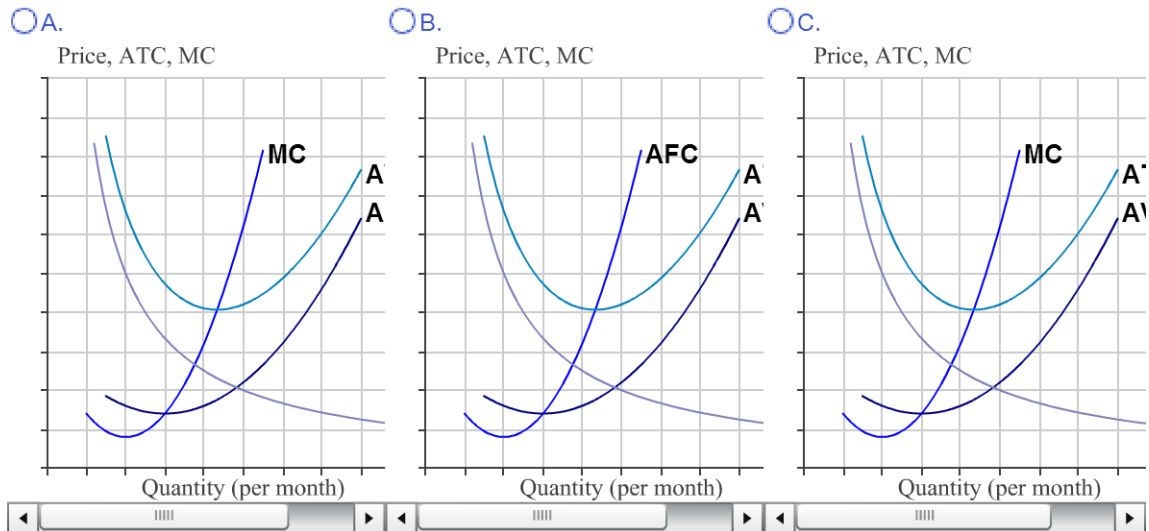
2,000

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16. Which of the graphs represents the correct relationship among the cost curves?



Answer: C

17. When the total product function begins to increase at a decreasing rate,

- ☐ A. marginal cost is rising.
☐ B. marginal product is falling.
☐ C. the law of diminishing returns has set in.
☐ D. All of the above.

Answer: D

18. When marginal cost is falling

- ☐ A. marginal product is at a maximum.
☐ B. marginal product must be rising.
☐ C. marginal product is at a minimum.
☐ D. marginal product must be falling.

Answer: B

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19. The shape of the short-run cost curves are the result of

- ☐ A. the law of diminishing returns.
- ☐ B. economies of scale.
- ☐ C. falling profits.
- ☐ D. diseconomies of scale.

Answer: A

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20.

Labor Input	Total Product	Marginal Product	Average Product
1	10	10	10.0
2	21	11	10.5
3	33	12	11.0
4	44	11	11.0
5	54	10	10.8
6	63	9	10.5
7	71	8	10.1

1.) Using the 4-point curved line drawing tool, the marginal product curve, and label this line 'MP'.

2.) Using the 4-point curved line drawing tool, the average product curve, and label this line 'AP'.

(For your control points, use 1, 3, 4, and 7 units of labor.)

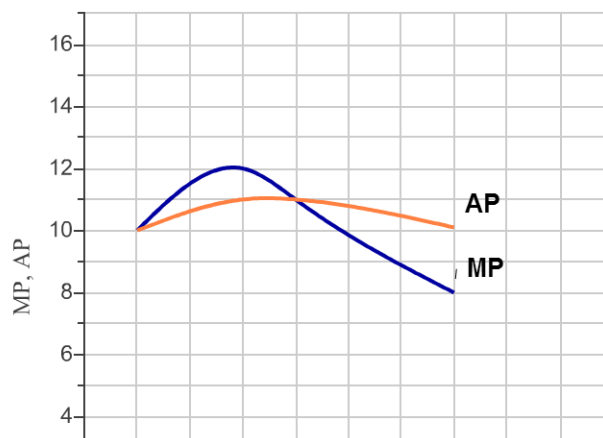
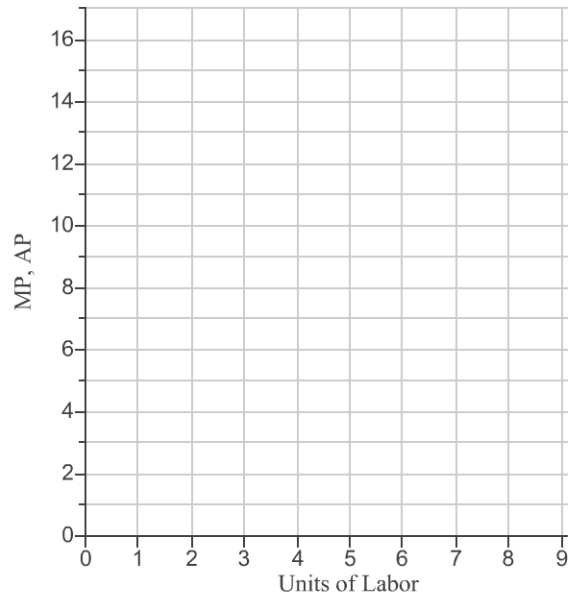
Carefully follow the instructions above, and then draw the required objects.

The marginal cost curve will intersect the average

variable cost curve at its minimum point, where

minimum
maximum

occurs when total product equals units.



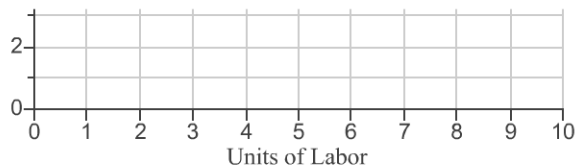
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20.
(cont.)

Answers



minimum

44

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21. The following table shows the daily relationship between the number of workers and output (Q) for a small factory in the short run, with capital held constant. Each worker costs \$150 per day, and the firm has fixed costs of \$75 per day. Calculate total cost (TC), marginal cost (MC), and average total cost (ATC). (*Round your answers to two decimal places.*)

Workers	Q	TC	MC	ATC
0	0		--	--
1	10			
2	22			
3	35			
4	46			
5	51			

Answers

75.00

225.00

15.00

22.50

375.00

12.50

17.05

525.00

11.54

15.00

675.00

13.64

14.67

825.00

30.00

16.18

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22. An electricity-generating company confronts the following long-run average total costs associated with alternative plant sizes. It is currently operating at plant size **G**.

The firm's minimum efficient scale occurs at .

If damage caused by a powerful hurricane generates a reduction in the firm's plant size from its current size to **B**, would there would

be a movement along the firm's long-run average total cost curve?

Answers E

leftward

Plant Size	Average Total Cost (\$)
A (<i>smallest</i>)	2,000
B	1,800
C	1,600
D	1,550
E	1,500
F	1,500
G (<i>largest</i>)	1,500

23. Which of the following statements describes a firm's long-run average cost curve?

- ☐ A. A U-shaped curve that represents the minimum unit cost of producing any given rate of output.
- ☐ B. A U-shaped curve that represents the lowest point on a series of short-run total cost curves.
- ☐ C. A U-shaped curve that represents the lowest point on a series of short-run variable cost curves.
- ☐ D. A U-shaped curve that represents the minimum point on a series of marginal cost curves.

Answer: A

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24. Which of the following is true about the long-run average cost curve?

- ☐ A. The long-run average cost curve is the envelope of the firm's short-run average cost curves.
- ☐ B. The long-run average cost curve is the locus of points of the minimum points of the firm's short-run average variable cost curves.
- ☐ C. The long-run average cost curve is the locus of points of the minimum points of the firm's short-run average total cost curves.
- ☐ D. The shape of the long-run average cost curves is determined by the law of diminishing marginal returns.

Answer: A

25. The minimum possible short-run average costs are equal to long-run average costs when

- ☐ A. the plant is producing at its short-run minimum point.
- ☐ B. short-run and long-run costs are declining.
- ☐ C. production is at any point on the long-run average cost curve.
- ☐ D. the long-run curve is at a minimum point.

Answer: D

26. In economics, the planning horizon is defined as

- ☐ A. the long run, during which all inputs are variable.
- ☐ B. the longest time period over which the firm can make decisions.
- ☐ C. 10 years.
- ☐ D. the period of time for which technology is fixed.

Answer: A

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27. For each example below, identify which statement is *not* characteristic of a perfectly competitive industry.

☐ A. One firm produces a large portion of the industry's total output.

☐ B. There are many firms in the industry.

☐ C. Their products are indistinguishable.

☐ D. Firms can easily exit and enter the industry.

☐ A. There are many buyers and sellers in the industry.

☐ B. Consumers have equal information about the prices of firms' products.

☐ C. The products differ slightly in quality from firm to firm.

☐ D. Many diners compete in a city.

☐ A. Many taxicabs compete in a city.

☐ B. The city's government requires all taxicabs to provide identical service.

☐ C. Taxicabs are virtually identical, and all drivers must wear a designated uniform.

☐ D. The government also limits the number of taxicab companies that can operate within the city's boundaries.

Answers A
 C
 D

28. Which of the following is *not* one of the assumptions of a perfectly competitive market?

☐ A. Better information for producers than consumers.

☐ B. Homogeneous product.

☐ C. Free entry and exit.

☐ D. Large number of buyers and sellers.

Answer: A

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29. The perfectly competitive firm is said to be a
- ☐ A. price participant — it decides the price together with other firms.
 - ☐ B. price leader — it changes its price and other firms follow.
 - ☐ C. price maker — it sets market prices.
 - ☐ D. price taker — it takes the price given by the market.

Answer: D

30. Firms in perfectly competitive industries will eventually have no customers if they set their prices above the competitive price.
- ☐ A. True
 - ☐ B. False

Answer: A

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31. Consider the market for DVD movie rentals, which is perfectly competitive. The market supply curve slopes upward, the market demand curve slopes downward, and the equilibrium rental price equals \$3.50. Indicate how each of the following events will have an effect on the **market** clearing price and on the demand curve faced by the **individual** rental store.

People's tastes change in favor of going to see more movies at cinemas with their friends and family members.

Market Clearing Price

decreases
increases

Firm's Demand Curve

shifts up
shifts down

National DVD-rental chains open a number of new stores in this market.

Market Clearing Price

increases
decreases

Firm's Demand Curve

shifts down
shifts up

There is a significant increase in the price of downloading movies on the Internet.

Market Clearing Price

decreases
increases

Firm's Demand Curve

shifts up
shifts down

Answers **decreases**
 shifts down
 decreases
 shifts down
 increases
 shifts up

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32. Consider the diagram at right, which applies to a perfectly competitive firm, which at present faces a market clearing price of \$20 per unit and produces 10,000 units of output per week.

What is the firm's current average revenue per unit?

\$.

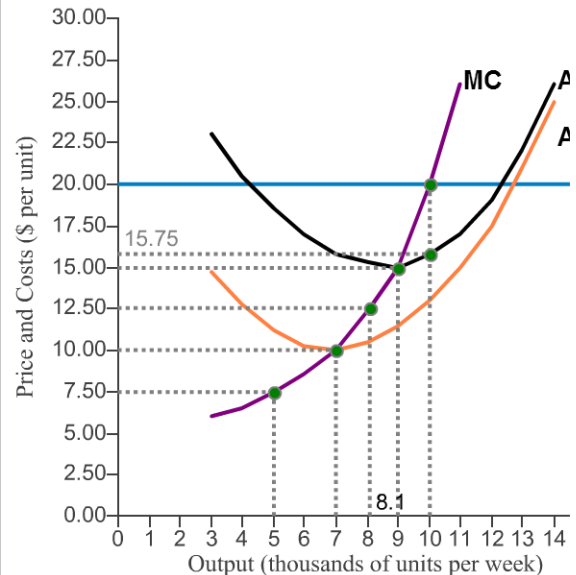
What are the present economic profits of this firm?

\$. (Remember that quantity is in thousands of units.)

Is the firm maximizing economic profits?

- ☐ A. Yes, they have economic profits.
- ☐ B. No, they are producing where the price equals the marginal cost.
- ☐ C. It is impossible to be sure with the given information.
- ☐ D. Yes, they are producing where the marginal revenue equals the marginal cost.
- ☐ E. Yes, they are producing where the price equals the average cost.

If the market clearing price drops to \$12.50 per unit the firm should



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32. (cont.)
- ☐ A. continue to produce since the loss is less than the fixed cost.
 - ☐ B. continue to produce since the firm can pay some of its fixed cost.
 - ☐ C. continue to produce since the price exceeds the average variable cost.
 - ☐ D. All of the above.
 - ☐ E. shut down since the price is less than the average total cost.

If the market clearing price drops to \$7.50 per unit the firm should

- ☐ A. shut down since the loss is more than the fixed cost.
- ☐ B. shut down since the firm can pay none of its fixed cost.
- ☐ C. shut down since the price is less than the average variable cost.
- ☐ D. All of the above.
- ☐ E. continue to operate at a loss until the firm can exit the market.

Answers 20

42,500

D

D

D

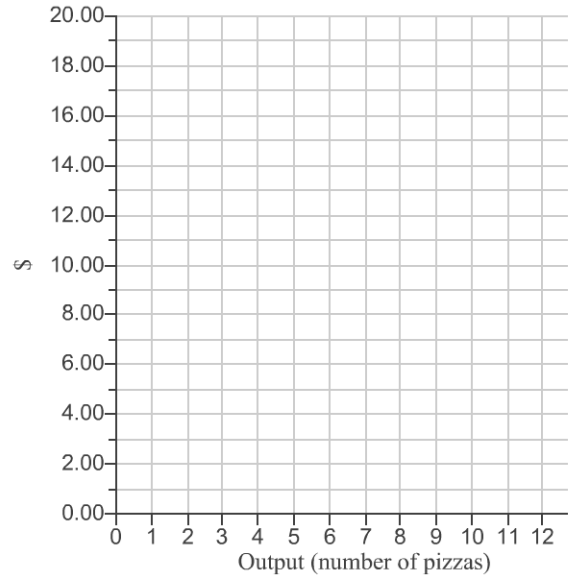
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Total Hourly Output & Sales of Pizzas	Total Hourly Cost (\$)	Total Revenue (\$)	Total Economic Profit (\$)	MC (\$)
0	2	0	0	XXX
1	6	8	2	5
2	8	16	8	2
3	9	24	15	1
4	11	32	21	2
5	15	40	25	4
6	21	48	27	6
7	29	56	27	8
8	39	64	25	10
9	51	72	21	12
10	67	80	13	14



Based on the information in the table above,

1.) Using the point drawing tool, plot two marginal cost (MC) points for an output level of 4 and 1 units. Do *not* label these points.

2.) Using the line drawing tool, draw the marginal cost from an output level of 4 to 10 units. Properly label this line.

3.) Using the line drawing tool, draw the marginal revenue line. Properly label this line.

Carefully follow the instructions above, and *or* draw the required objects.



The profit maximizing output level is pizzas

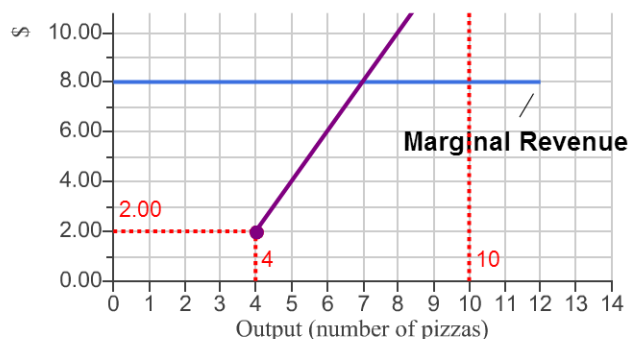


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33.
(cont.)



Answers

7

34. Suppose that a firm in a perfectly competitive industry finds that at its current output rate, marginal revenue exceeds the minimum average total cost of producing any feasible rate of output. Furthermore, the firm is producing an output rate at which marginal cost is less than the average total cost at that rate of output.

Is the firm maximizing its economic profits?

- ☐ A. Yes, marginal revenue exceeds the minimum average total cost of producing any feasible rate of output.
- ☐ B. It is impossible to draw a conclusion from the given information.
- ☐ C. No, the firm is producing where average total cost are a minimum.
- ☐ D. No, if the firm was maximizing its economic profits the marginal cost would not be less than the average total cost at that rate of output.
- ☐ E. No, the marginal cost is less than the average cost, so the average cost must be declining at the output level where the firm is producing.

Answer: D

35. The demand curve for the perfectly competitive firm is

- ☐ A. perfectly inelastic.
- ☐ B. perfectly elastic.
- ☐ C. elastic at lower output levels, then unit elastic, and then inelastic at higher output levels.
- ☐ D. unit elastic.

Answer: B

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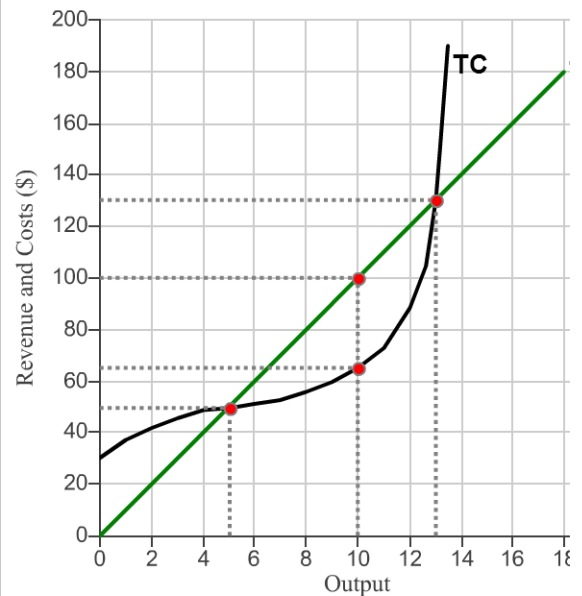
36. The demand curve for the perfect competitor is horizontal because

- ☐ A. the market dictates each firm's price.
- ☐ B. demand is perfectly inelastic.
- ☐ C. the firm is a price maker.
- ☐ D. None of the above.

Answer: A

37. In the figure to the right, the firm should produce

- ☐ A. 5 or 13 units since the firm breaks even.
- ☐ B. any output level between 5 and 13.
- ☐ C. 10 units since economic profits are the greatest.
- ☐ D. 10 units since this is the only output level with accounting profits.



Answer: C

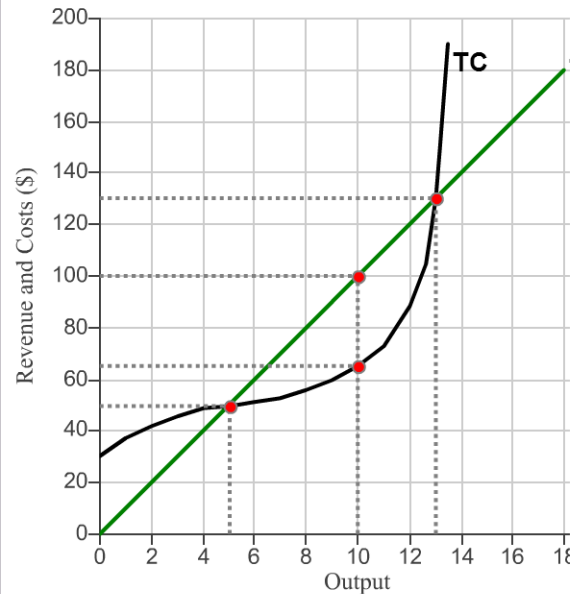
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38. In the figure to the right, at the output level between 5 units and 13 units

- ☐ A. the firm's accounting profits are negative.
- ☐ B. the firm is breaking even.
- ☐ C. total revenue equals total costs.
- ☐ D. the firm's economic profits are positive.



Answer: D

39. Suppose that a perfectly competitive firm faces a market price of \$5 per unit, and at this price the upward-sloping portion of the firm's marginal cost curve crosses its marginal revenue curve at an output level of 1,500 units. If the firm produces 1,500 units, its average variable costs equal \$5.50 per unit, and its average fixed costs equal \$0.50 per unit.

What is the firm's profit-maximizing (or loss-minimizing) output level? .
(Enter your response as a whole number — include the minus sign if necessary.)

What is the amount of its economic profits (or losses) at this output level? \$.
(Enter your response as a whole number — include the minus sign if necessary.)

Answers 1,500
- 1,500

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40. Profits are maximized for the perfectly competitive firm when the firm produces the quantity where

- ☐ A. $MC = MR$.
- ☐ B. total revenue exceeds total cost by the greatest amount.
- ☐ C. $P = MC$.
- ☐ D. All of the above.

Answer: D

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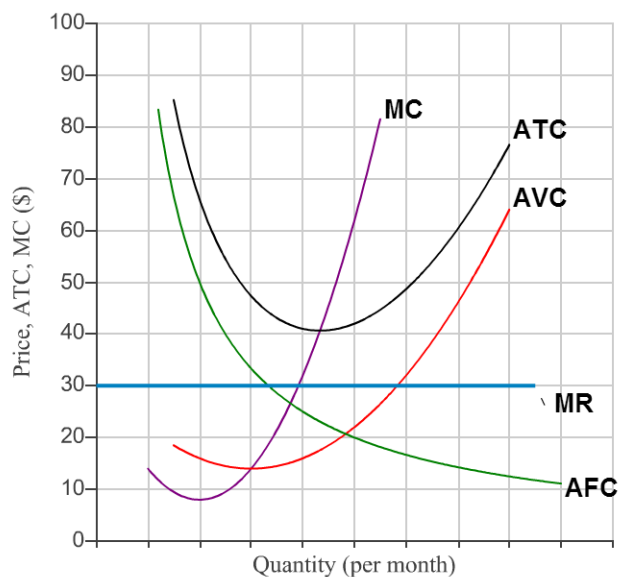
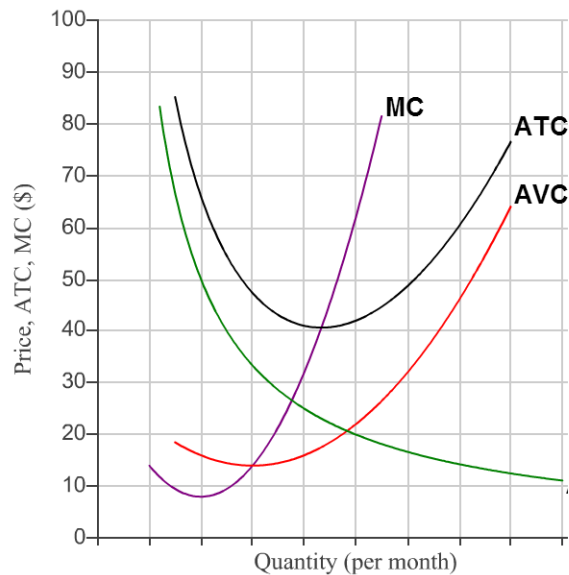
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41. Suppose that the market price in a perfectly competitive industry is \$30 per unit.

Using the line drawing tool, plot a possible marginal revenue line given this price. Label this line 'MR'.

Carefully follow the instructions above, and only draw the required objects.



Answer:

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42. The following table shows the total cost schedule for a perfectly competitive firm. The current price in this industry is \$8. Fill in the marginal revenue and marginal cost columns of the table. *(Enter your response as an integer.)*

Output (units) (Q)	Total Cost (TC)	Marginal Revenue (MR)	Marginal Cost (MC)
0	20	--	--
1	22		
2	26		
3	32		
4	40		
5	50		

A profit-maximizing firm will produce units. *(Enter your response as an integer.)*

Answers 8
 2
 8
 4
 8
 6
 8
 8
 8
 10
 4

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43. The table at right represents the hourly output and cost structure for a local pizza shop. The market is perfectly competitive. Total costs include all implicit opportunity costs. Suppose the market price drops to only \$5 per pizza.

Total Hourly Output and Sales of Pizzas	Total Hourly Cost (\$)
0	5
1	9
2	11
3	12
4	14
5	18
6	24
7	32
8	42
9	54
10	68

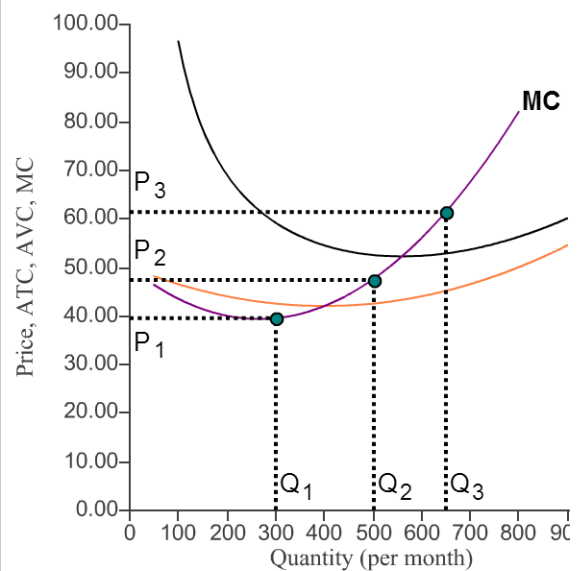
In the short run, this pizza shop

- ☐ A. maximize its average profits by shutting down.
☐ B. should continue to make pizzas but raise the price.
☐ C. should continue to make pizzas at the \$5 price.
☐ D. minimize its economic loss by shutting down.

Answer: C

44. In the graph, if price is P_1 , the profit-maximizing rate of output is

- ☐ A. zero.
☐ B. Q_1 .
☐ C. Q_2 .
☐ D. Q_3 .



Answer: A

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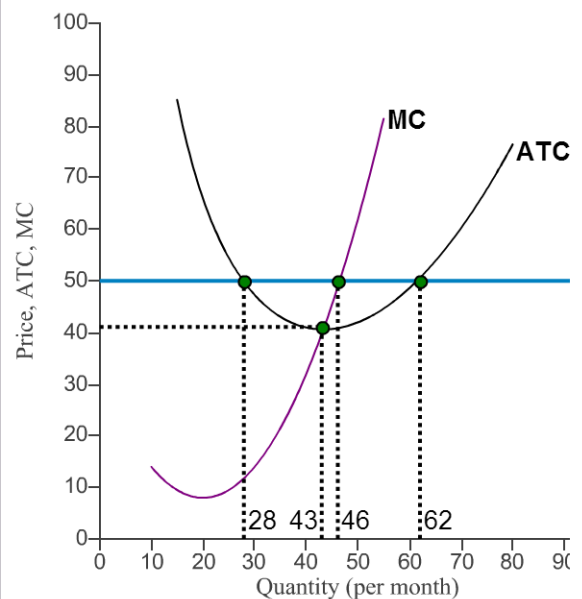
45. If a perfectly competitive firm sells the product for a profit-maximizing price of \$4.76 and has average total cost per unit of \$5.16, in the short run

- ☐ A. this firm must hope the market price rises soon or exit the industry.
- ☐ B. this firm should shut down if \$4.76 is less than minimum AVC.
- ☐ C. this firm is losing money.
- ☐ D. All of the above.

Answer: D

46. In the figure shown at the right, what is the profit maximizing output and price in the short run?

- ☐ A. 46, \$50
- ☐ B. 43, \$41
- ☐ C. 28, \$50
- ☐ D. 62, \$50



Answer: A

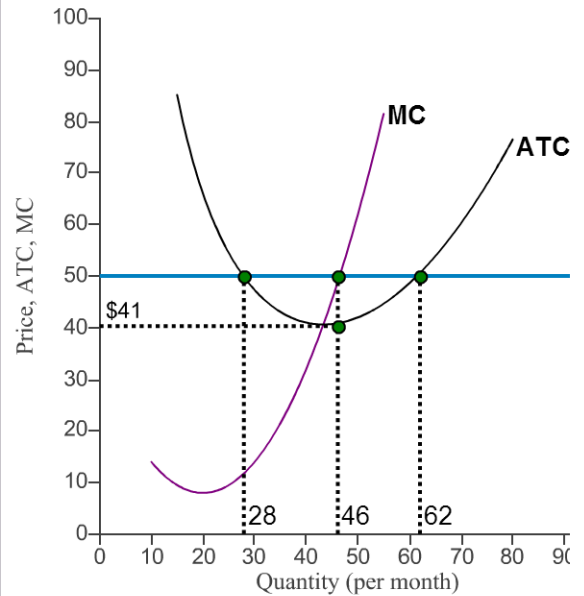
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47. In the figure shown at the right, what is the profit at the profit maximizing output and price?

- ☐ A. \$414
☐ B. \$1,886
☐ C. \$0
☐ D. \$2,300



Answer: A

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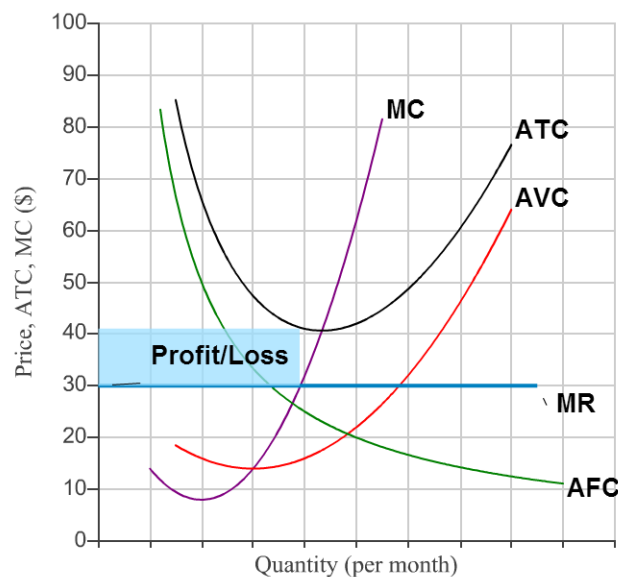
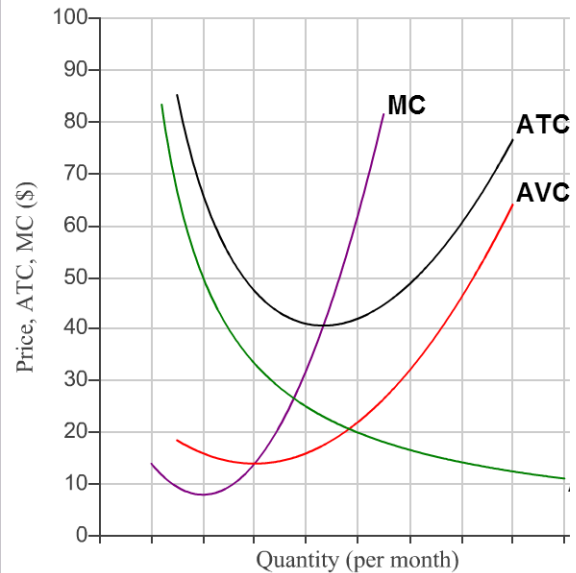
48. Suppose that the market price in a perfectly competitive industry is \$30 per unit.

1.) Using the line drawing tool, draw the marginal revenue line given this price and label your line MR.

2.) Using the rectangle drawing tool, draw a rectangle showing the area of profit or loss of the firm and label it P/L.

Carefully follow the instructions above, and only draw the required objects.

The firm is operating at an economic loss
 in the short run.



Answers

loss

Student: _____
Date: _____
Time: _____

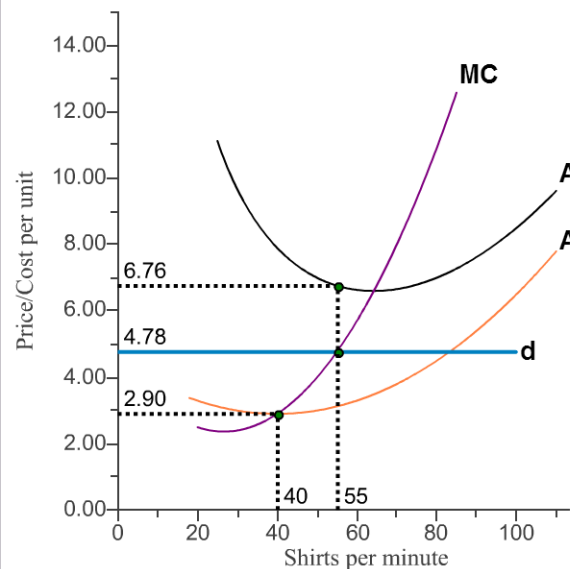
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49. The graph to the right depicts the per unit cost curves and demand curve facing a shirt manufacturer in a competitive industry.

How much profit is this firm making per minute?

\$ per minute. (Round your answer to the nearest penny. Include a minus sign for losses.)



Answer: - 108.90

50. The lowest profit a firm should ever make in the short run is

- ☐ A. zero accounting profits.
☐ B. zero economic profits.
☐ C. a normal return.
☐ D. the losses associated with the fixed costs of the firm.

Answer: D

51. A firm will continue to operate in the short run, even at an economic loss, as long as

- ☐ A. P is greater than minimum AVC .
☐ B. $MC = P$.
☐ C. $MR = MC$.
☐ D. $ATC > P$.

Answer: A

Student: _____
Date: _____
Time: _____

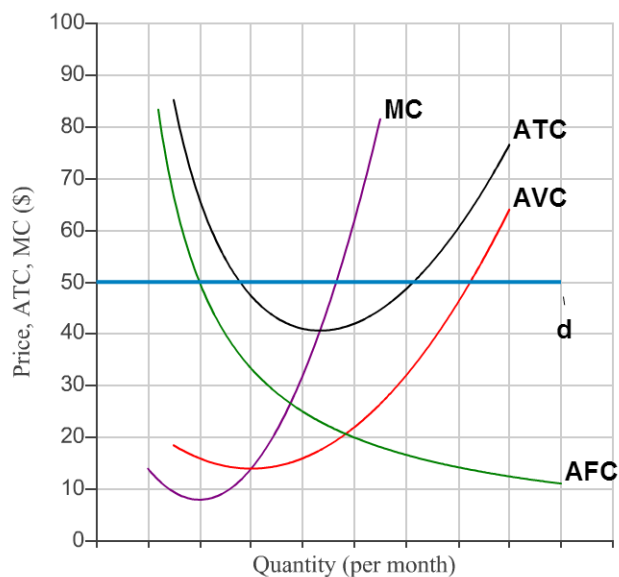
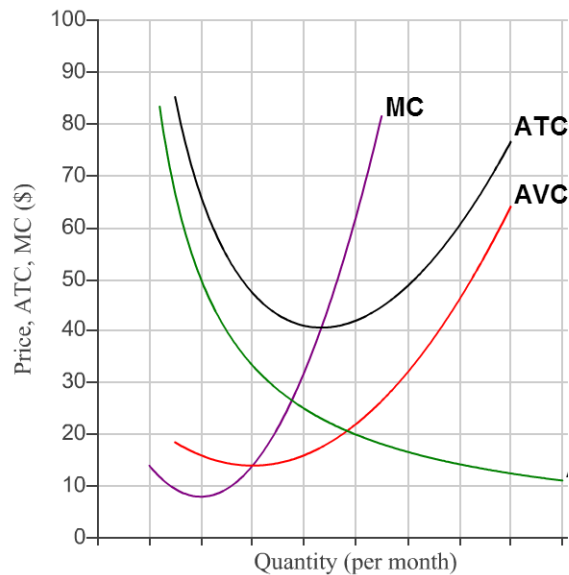
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Assignment: Assignment 4

52. The graph to the right depicts the average cost curves and the marginal cost curve for a typical firm in a competitive industry.

Using the line drawing tool, draw the firm's demand curve at a market price such that the firm is making an economic profit. Label your curve 'd'.

Carefully follow the instructions above, and only draw the required objects.



Answer:

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 Time: _____

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53. The graph to the right depicts the average cost curves and the marginal cost curve for a typical firm in a competitive industry.

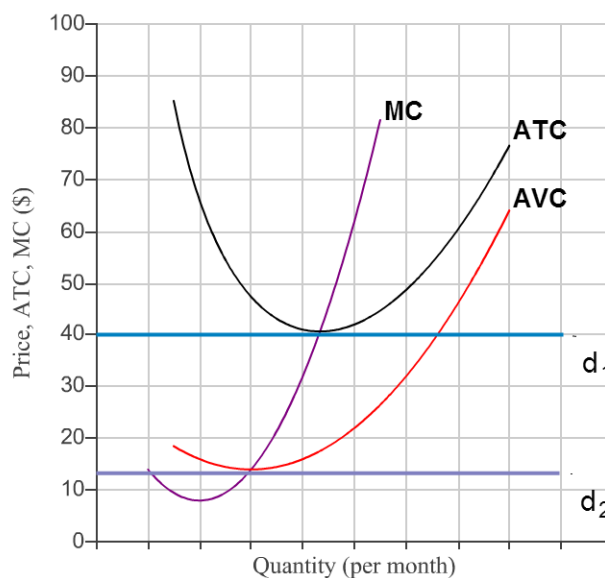
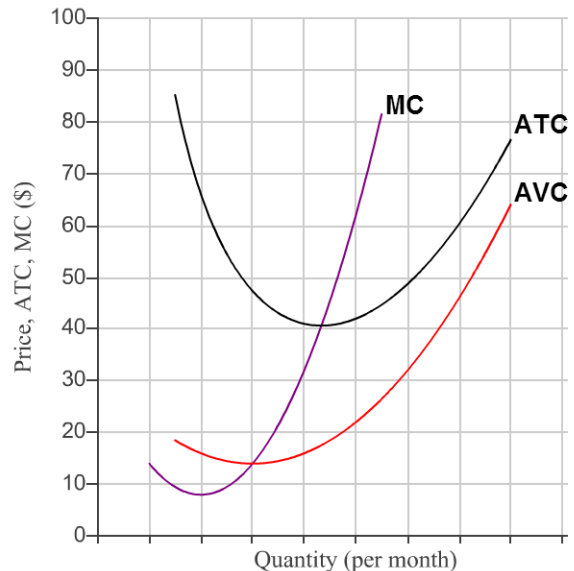
1.) Using the line drawing tool, draw the firm's demand curve at a market price such that the firm is breaking even. Label your curve 'd₁'.

2.) Using the line drawing tool, draw the firm's demand curve at a market price such that the firm is at its shutdown price. Label your curve 'd₂'.

Carefully follow the instructions above, and only draw the required objects.

When the firm is at its shutdown price,

- ☐ A. it has losses equal to fixed costs.
- ☐ B. it will not shutdown if the price falls.
- ☐ C. it is making zero economic profit.
- ☐ D. None of the above.



Answers

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53.

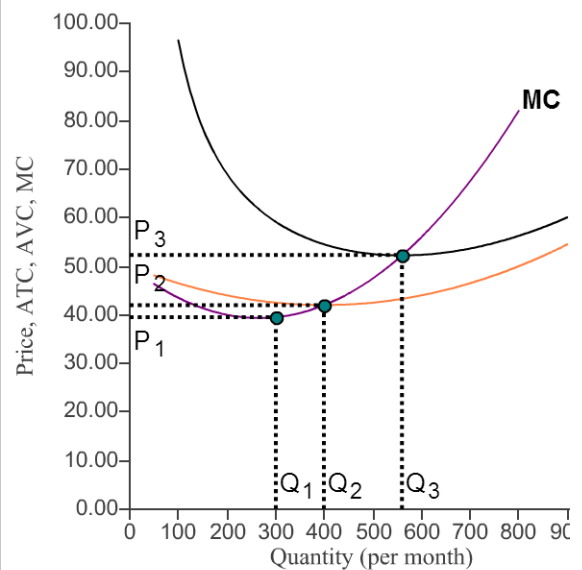
A

(cont.)

54.

In the graph, the firm's supply curve is

- ☐ A. the average total cost curve.
- ☐ B. the marginal cost curve that lies above P_1 .
- ☐ C. the average variable cost curve.
- ☐ D. the marginal cost curve that lies above price P_2 .



Answer: D

55.

In a competitive market, positive economic profits act to

- ☐ A. drive potential competitors away from the industry.
- ☐ B. attract new entrants into the industry.
- ☐ C. prevent reinvestment on the part of firms within the industry.
- ☐ D. signal resource owners elsewhere not to invest their capital in this industry.

Answer: B

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56. When the perfect competitor earns less than normal profits in the long run, the firm will

- ☐ A. increase production.
- ☐ B. exit the industry.
- ☐ C. set a new price.
- ☐ D. decrease production.

Answer: B

57. The perfectly competitive firm in long-run equilibrium produces a level of output such that

- ☐ A. $AFC + AVC = ATC$.
- ☐ B. P exceeds ATC by the greatest amount.
- ☐ C. there is entry of new firms into the industry.
- ☐ D. $P = MC = MR = \text{short-run } ATC = \text{long-run } ATC$.

Answer: D

58. A perfectly elastic long-run supply curve

- ☐ A. indicates that input prices do not change when firms enter or exit the industry.
- ☐ B. is not possible when the firm's demand curve is also perfectly elastic.
- ☐ C. indicates that short-run profits are zero.
- ☐ D. is derived from the horizontal summation of the firms' long-run supply curves.

Answer: A

59. In long-run equilibrium, the perfectly competitive firm makes

- ☐ A. positive economic profits.
- ☐ B. negative economic profits.
- ☐ C. positive accounting profits but negative economic profits.
- ☐ D. zero economic profits.

Answer: D

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 Time: _____

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60. The graph to the right depicts the per unit cost curves and demand curve facing a shirt manufacturer in a competitive industry.

How much profit is this firm making per minute?

\$ per minute. (Round your answer to the nearest penny.)

In the long run, firms will this

industry until there is a tendency for the firms

in the industry to be making

economic profit.

Answers – 108.90

exit

zero

